Exact public kernel line and constrained rank repair

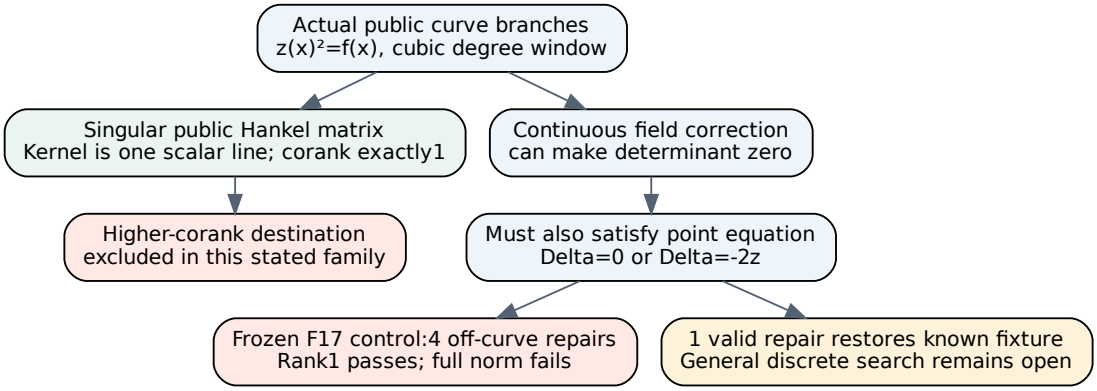
2026-11-13. Source`4b6ed788d`. Unsigned evidence [pcc:unverified].

Exact geometry. Over a field, let $|S| = 2a$, $a \geq 2$, $\deg f = 3$, and $z(x)^2 = f(x)$ on every actual node. Let M be the public barycentric Hankel matrix of size $a - 1$. If $\det M = 0$, then

$$\operatorname{rank}(M) + 1 = a - 1, \qquad Mb = 0 \iff \exists t \in K,\ b = tu$$

for every nonzero actual kernel vector u. Numerical corank is derived, not assumed. No successful A/B, primitive-polynomial premise, normalization, square-root or rank oracle is supplied [pcc:unverified].

Direct proof. Actual reconstruction gives bounded A_0, B_0, A_1, B_1. Shared node branches imply $G \mid A_0B_1 - A_1B_0$, whose degree is below $\deg G = 2a$; hence the cross product is zero. The derived nonzero norm identities then give $c_0B_1^2 = c_1B_0^2$. Leading coefficients construct the scalar ratio, and factoring a difference of squares yields $B_1 = CtB_0$ without a root oracle. Injectivity of ofFn gives the vector line; ordinary rank-nullity gives corank one. Zero vectors are handled separately [pcc:unverified].



Curve constraint. For every field,

$$(z + \Delta)^2 = z^2 \quad \iff \quad \Delta = 0 \text{ or } \Delta = -2z.$$

Factor $\Delta(2z + \Delta)$. In characteristic2 the alternatives coincide; for prime-curve inputs $p > 3$, $z \neq 0$, the only nontrivial correction is the opposite branch. A field-valued determinant repair must satisfy this as well as singularity [pcc:unverified].

A larger-corank target is excluded for this stated public family. This is not a global ECDLP lower bound, a no-go result for all discrete searches, or a map from the earlier W-jet proposal [pcc:unverified].

Fixed repair controls and remaining search obligation

Over F17 use $f = X^3 + 2X + 2$ and supplied points $(0,6),(3,1),(5,1),(6,3),(10,11),(13,10)$. Initially $M = \left(\begin{smallmatrix} 9 & 14 \\ 14 & 3 \end{smallmatrix}\right)$, $\det M = 1$. For $v = (1, x)$ set $h = w_x v^T \operatorname{adj}(M) v$. If $h \neq 0$, choose $\Delta = -\det M/h$ and rebuild the matrix. Record $h = 0$ without dividing. All17 field corrections are tested for that frozen zero direction [pcc:unverified].

x	h	Δ	new y	Actual curve / full norm
0	12	7	13	fails / fails; target branch changes
3	0	–	–	no continuous repair in this direction
5	11	3	4	fails / fails
6	16	1	4	fails / fails
10	6	14	8	fails / fails
13	6	14	7	passes / passes

All five repaired matrices have determinant0 and rank1. Reconstruct every kernel vector to A/B and check repaired branch rows, degree bounds, the actual curve equation and full norm. The four off-curve repairs fail the full norm despite rank1. The one valid repair restores the known positive fixture. These are fixed correctness cases, not independent trials, a sign-search success rate or a runtime result [pcc:unverified].

With target branch at $x = 0$ fixed, only four repairs are eligible: three fail the curve/norm and one passes; the same one zero direction remains. These are two views of the same cases. A separate finite predicate check covers all289 pairs (z, Δ) over F17 [pcc:unverified].

Validation. Eleven new statements in NormBranchRepair. Build001 fails on two interface details;002 passes and prints only standard Lean foundations. Whole-study source audit and fresh import-inclusive replay have exact receipts in checks.json/fresh-status.json. Eleven native tests, strict clippy, release build001 and one write-once artifact invocation001 pass; failed source log/exit retained [pcc:unverified].

Kernel line: `PCC-20261113-98d020`.

Actual corank: `PCC-20261113-a3591d`.

Exact branch constraint: `PCC-20261113-59cebf`.

Original native artifact: `PCC-20261113-49efb0`.

All unsigned. Default-PATH source-only checks are not_covered, separately from explicit-toolchain kernel checks. Pinned Lean4.35.0, CryptoLean `148f5dc389b`, Mathlib`cfc55fc58714`. The README contains the full 11-statement proof map, artifact hash and source-linked original evidence.

Next action. A candidate must implement genuinely discrete branch selection or a different public factor-base construction with fewer counted attempts, while keeping target, norm/divisor/scalar checks and total $\exp(o(\log r))$ bit work. Continuous repair and higher corank do not supply that result. No general existence/nonexistence theorem follows [pcc:unverified].

Sources. Actual public Hankel construction and pinned polynomial/ rank-nullity declarations; previously opened Li–Wan–Zhang1501.01138 Proposition4.1 and Cheng cs/0507026 principal-divisor discussion. The proof is direct algebra with no paper statement encoded as an axiom. KB unavailable: MCP absent, wrapper has no executable. No novelty/exhaustive claim, paid run, statistical yield or performance promotion.